Financial Derivatives
ES32079

Lecture 1

Introduction to Financial Derivatives

Aliya Kenjegalieva

About me

Dr Aliya Kenjegalieva Lecturer in Economics
Director of Internationalisation & Study Abroad Program

My role in this unit

- Expect clear guidance, worked examples, and constructive feedback
- Use support early (Q&A forums, emails, office hours)

Contact & support

- Email: `ak3085@bath.ac.uk`
- Office: 3 East 3.10
- Office hours: Thursdays 15:15-16:00
- Reply policy: I typically respond within 2-3 working days; please avoid duplicate emails.

Timetable

Type	**Day & Time**	**Room**	**Week Pattern**
Lecture	Wednesdays, 08:15-10:05	2E 3.1	Weeks 1-5, 7-11
Seminar 1	Thursdays, 14:15-15:05	CB 3.1	Weeks 2-5, 7-11
Seminar 2	Thursdays, 16:15-17:05	1W 2.101	Weeks 2-5, 7-11

Key People

- **Unit Convenor / Lecturer:** Dr Aliya Kenjegalieva
  Email: ak3085@bath.ac.uk
- **Economics UG Admins:** economics-ug-admin@bath.ac.uk – Enrolment, Moodle access, exam queries
- **Director of Teaching:** Dr Atisha Ghosh – ag880@bath.ac.uk
- **Director of Studies:** Dr Maria Sarigiannidou – ms962@bath.ac.uk

A £10 Million Problem

The situation

A UK airline must buy \$10 million of fuel in three months.
Today: \$1 = £0.78
What happens if the dollar becomes more expensive?

What should the airline do?

1. Buy the dollars today
2. Agree today on a future exchange rate
3. Buy the right, *but not the obligation*, to exchange later
4. Do nothing

What Will This Unit Help You Do?

By the end of this unit, you should be able to:

- design derivatives to transfer or manage financial risk;
- represent their cash flows mathematically;
- identify arbitrage opportunities;
- price derivatives using replication and no-arbitrage;
- evaluate the limitations of pricing models.

Our central question

How can we place a price today on a payoff that depends on an uncertain future?

Our Learning Journey

1. **Contracts and risk**
   What derivatives are and how firms use them
2. **No-arbitrage pricing**
   Interest rates, bonds, forwards, and futures
3. **Options and non-linear payoffs**
   Option properties and trading mechanisms
4. **Pricing under uncertainty**
   Replication and binomial models
5. **Extensions and limitations**
   Credit risk and exotic options

Recommended Reading and Resources

Main Textbook

- Hull, J.C. *Options, Futures and Other Derivatives* - any edition

Additional References

- Bailey, R.E. *The Economics of Financial Markets*
- Bodie, Z., Kane, A., and Marcus, A.J. *Investments*
- Brealey, R.A., Myers, S.C., and Allen, F. *Principles of Corporate Finance*

Other Resources

- Lecture slides and weekly materials on Moodle
- Relevant journal articles, case studies, and financial market data (provided during the semester)

Assessment Overview

- The unit is assessed by a **100% final examination**.
- The examination will take place during the **January exam period**.
- Detailed information on exam length, structure, and permitted materials will be provided as soon as it is confirmed by the Department.
- Mock exam questions will be provided and discussed during the semester to help you prepare.
- *All of the seminar questions are similar to the exam questions, so make sure you understand them. From Week 6 onwards, the structure of the seminar problem sets will replicate the exam format.*

Structure of this lecture

Today we cover the following topics:

- Introduction to financial derivatives

Reference: Hull chapter 1.

What Do These Contracts Have in Common?

Consider three contracts

- A contract fixing the price of oil delivered in three months
- A contract paying £100 if the FTSE 100 exceeds a specified level
- A contract paying a farmer if rainfall falls below a threshold

Question: What determines the value of each contract?

Key idea

A **derivative** is a financial contract whose value depends on the value or outcome of an underlying asset, rate, index, or event.

Examples include forwards, futures, swaps, and options.

Why Use Derivatives?

Three decisions

1. An airline fixes the future price of fuel.
2. A trader uses an option to profit from an expected rise in volatility.
3. A dealer exploits inconsistent prices in two markets.

Match each decision to:

Hedging Speculation Arbitrage

The three principal uses

- **Hedging:** reducing or transferring an existing risk
- **Speculation:** taking risk in anticipation of a return
- **Arbitrage:** exploiting inconsistent prices

Where Are Derivatives Traded?

Exchange-traded

- Standardised contracts
- Transparent market prices
- Clearinghouse involvement
- Usually high liquidity

Over-the-counter

- Privately negotiated
- Can be customised
- Often used by institutions
- May be centrally cleared

Quick decision

A company needs a contract tailored to an unusual exposure. Which market is more likely to meet its needs and what might it give up?

Before 2008: Risk Was Difficult to See

A network of bilateral promises

Many OTC derivatives were negotiated privately between banks and other institutions.

- Contracts could be tailored to the parties' needs
- Banks commonly acted as dealers and market makers
- Counterparty exposures accumulated across institutions
- Reporting and price transparency were limited
- Many contracts were not centrally cleared

The systemic-risk problem

If one major counterparty failed, losses could spread through a network of contracts whose total exposure was difficult for participants and regulators to observe.

After the Crisis: Making OTC Markets Safer

The post-crisis reforms

Depending on the contract, counterparty, and jurisdiction:

- eligible standardised OTC derivatives may have to be cleared through a central counterparty;
- derivative transactions are reported to trade repositories;
- uncleared contracts may face margin and risk-management rules;
- some standardised contracts trade on organised platforms.

Quick check

Which reform primarily addresses each problem?

1. Regulators cannot observe accumulated exposures.
2. One counterparty may fail before paying what it owes.
3. A customised contract cannot be centrally cleared.

What Does a Central Counterparty Do?

Without central clearing

Bank A and Bank B are directly exposed to each other's default.

With central clearing

The CCP becomes:

- the buyer to Bank A's sale; and
- the seller to Bank B's purchase.

Key implication

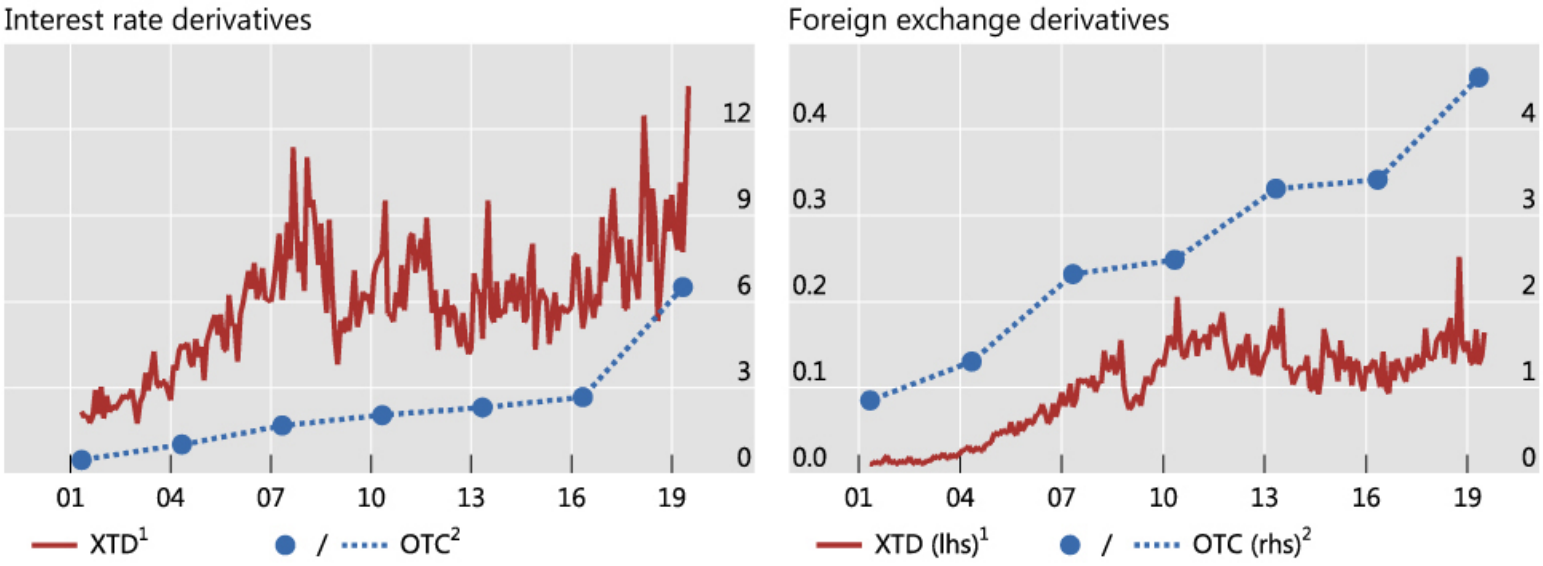
The original trade remains economically matched, but counterparty exposure is managed through the CCP's margin and default procedures.

Question: Does clearing eliminate risk *or* reorganise and manage it?

Most FX Derivative Trading Occurs OTC

OTC trading of interest rate and FX derivatives outpaced exchange trading

Daily average turnover, in trillions of US dollars — Graph A1



XTD = exchange-traded derivatives; OTC = over-the-counter derivatives.

[1] Turnover on exchanges worldwide, at monthly frequency. [2] Turnover in April, adjusted for local and cross-border inter-dealer double-counting. The dashed line shows a linear interpolation of data between Triennial Surveys.

Sources: Euromoney TRADEDATA; Futures Industry Association; The Options Clearing Corporation; BIS derivatives statistics.

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Source: BIS, Triennial Central Bank Survey, 2019. OTC observations occur at survey dates; dotted lines interpolate between them.

Most FX Derivative Trading Occurs OTC

Read the evidence

1. What exactly is being measured?
2. Which market dominates FX derivative trading?
3. Can turnover be interpreted as the amount of money at risk?

The Same Derivative Can Serve Different Purposes

Classify each position

1. An airline buys oil futures because it must purchase fuel in six months.
2. A trader buys oil futures because she expects oil prices to rise.
3. A dealer simultaneously buys and sells equivalent contracts quoted at inconsistent prices.

Purpose depends on the existing exposure

1. **Hedging**: offsets an existing business risk
2. **Speculation**: deliberately takes market risk
3. **Arbitrage**: exploits inconsistent prices

Can We Fix Today the Price Paid in Three Months?

The situation

A UK firm will need to buy one unit of an asset in three months.
The asset costs £100 today, but its future price is uncertain.

A forward contract

The firm agrees today to buy the asset in three months for a price of £105.

- The transaction occurs at a specified future date, T
- The delivery price, $K = 105$, is fixed today
- Both parties are obliged to complete the transaction

Every Forward Has Two Sides

Long forward

The long party agrees to:

- buy the underlying asset;
- at time T;
- for the delivery price K.

Short forward

The short party agrees to:

- sell the underlying asset;
- at time T;
- for the delivery price K.

Quick check

If the market price at maturity rises above K, which party benefits?

The Long Forward Payoff Is $S_T - K$

Suppose the delivery price is $K = 105$.

S_T	**Asset received**	**Payment**	**Long payoff**
£80	+80	−105	−25
£105	+105	−105	0
£130	+130	−105	+25

General result

At maturity, the payoffs are

$$\Pi_T^{\text{long}} = S_T - K, \qquad \Pi_T^{\text{short}} = K - S_T.$$

The Forward Rate Depends on Maturity

	Bid	Offer
Spot	1.5541	1.5545
1-month forward	1.5538	1.5543
3-month forward	1.5533	1.5538
6-month forward	1.5526	1.5532

Read the quotation

1. What is the spot rate?
2. What six-month rate can be fixed today?
3. Why might a firm prefer the forward rate to waiting?

Exchange-rate convention: USD per GBP.

A Treasurer Must Pay £1 Million in Six Months

The exposure

A US firm must pay a UK supplier £1 million in six months.
If sterling appreciates, the dollar cost of the payment will rise.

The hedge

The treasurer enters a long forward to buy £1 million at

$$K = 1.5532 \text{ USD per GBP}.$$

The dollar payment fixed today is

$$\$1{,}000{,}000 \times 1.5532 = \$1{,}553{,}200.$$

Question: Has the treasurer guaranteed a profit *or* guaranteed the dollar cost of the future payment?

What Is the Forward Worth at Maturity?

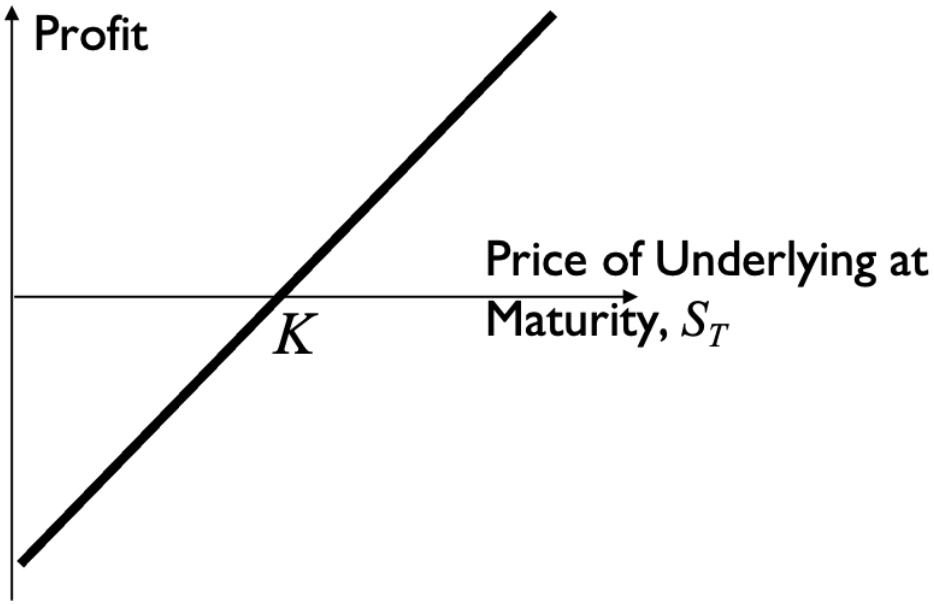
$$\Pi_T^{\text{long}} = 1{,}000{,}000(S_T - K), \qquad K = 1.5532.$$

Spot rate S_T	**Forward payoff**	**Interpretation**
1.4500	−\$103,200	Sterling depreciated
1.5532	\$0	Break-even
1.6500	+\$96,800	Sterling appreciated

General result

The long forward gains when $S_T > K$ and loses when $S_T < K$.

The Long Forward Gains When $S_T > K$



Reading the graph

The payoff is

$$\Pi_T^{\text{long}} = S_T - K.$$

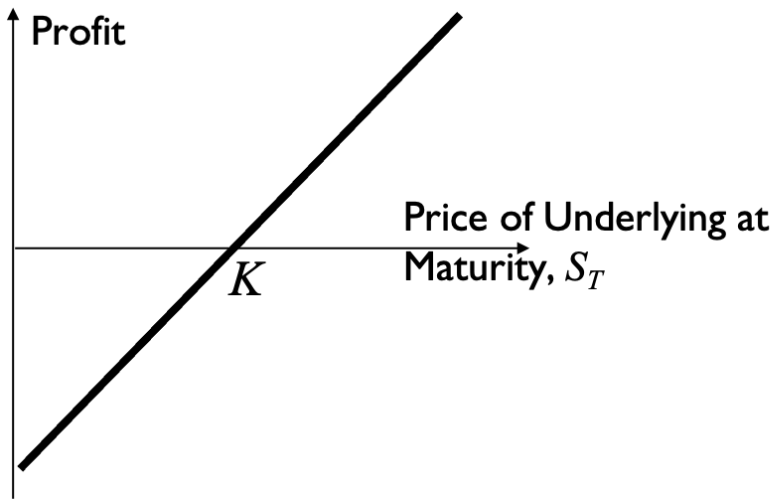
The line has slope $+1$: a £1 increase in S_T increases the payoff by £1 per unit.

Prediction

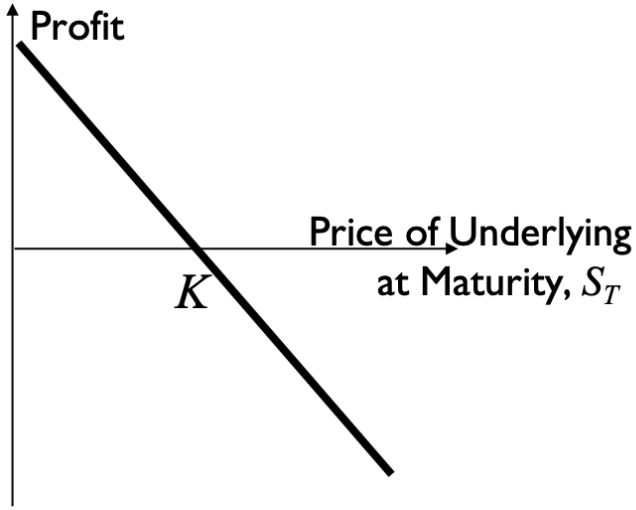
What should the short-forward payoff graph look like?

Long and Short Forward Payoffs

Long forward



Short forward



$$\Pi_T^{\text{long}} = S_T - K$$

$$\Pi_T^{\text{short}} = K - S_T$$

What do you notice?

$\Pi_T^{\text{long}} + \Pi_T^{\text{short}} = 0$: one party's gain is the other party's loss.

A Futures Contract Resembles a Forward - But Trades Differently

What remains the same?

A futures contract creates an obligation to buy or sell an underlying asset at a specified price and future date.

	Forward	**Futures**
Trading venue	Usually OTC	Exchange-traded
Contract terms	Customised	Standardised
Counterparty	Other party	Clearinghouse
Settlement	Mainly at maturity	Daily gains and losses
Margin	Contract-dependent	Margin account required

Key question

If long and short futures payoffs offset each other, why does the exchange require both parties to post margin?

Futures Gains and Losses Are Settled Daily

Example

A trader takes a long position in one futures contract.
Each contract covers 100 units, and the futures price rises from £50 to £53 during the day.

Daily gain

$$\text{Daily gain} = \text{contract size} \times \text{price change}$$

$$= 100(53 - 50) = 300.$$

General result

For a long position in N contracts, each covering q units,

$$\Delta V_t = Nq(F_t - F_{t-1}).$$

For a short position, the sign is reversed.

Margin Reduces the Risk of Non-Payment

- **Initial margin:** funds deposited when the position is opened
- **Variation margin:** daily gains or losses added to the account
- **Maintenance margin:** minimum permitted account balance
- **Margin call:** request to restore the account balance

Calculate the margin balance

A trader deposits initial margin of £2,000.
The futures position loses £600 on Day 1 and £500 on Day 2.

$$\text{Margin balance after Day 2} = ?$$

Answer

$$2{,}000 - 600 - 500 = 900.$$

If the maintenance margin exceeds £900, the trader receives a margin call.

Can You Read a Futures Position?

Three positions

1. Long 100 oz. of gold for December delivery at US$1,400 per oz.
2. Short £62,500 for March delivery at 1.5500 USD per GBP.
3. Short 1,000 barrels of oil for April delivery at US$90 per barrel.

For each position, identify:

1. the underlying;
2. the contract size;
3. the delivery month;
4. the agreed futures price;
5. whether the trader benefits from a price rise or fall.

Would You Pay to Keep a Choice Open?

Two contracts

You may acquire an asset in three months for £100.

1. You are **obliged** to buy it for £100.
2. You have the **right, but not the obligation**, to buy it for £100.

The key distinction

- A forward or futures contract creates an **obligation**.
- An option gives its buyer a **right**.
- The option buyer pays a **premium** for that right.

Question: Which contract is more valuable to its holder and why?

A Call Option Gives the Right to Buy

Call option

A call gives its holder the right, but not the obligation, to buy an underlying asset for the strike price K.

Exercise decision

Suppose $K = 100$.
Would you exercise the call if the market price at maturity is:

$$S_T = 70, \qquad S_T = 100, \qquad S_T = 130?$$

S_T	**Exercise?**	**Payoff**
£70	No	£0
£100	Indifferent	£0
£130	Yes	£30

$$C_T = \max(S_T - K, 0).$$

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S_T	**Exercise?**	**Payoff**
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£100	Indifferent	£0
£130	Yes	£30

$$C_T = \max(S_T - K, 0).$$

A Put Option Gives the Right to Sell

Put option

A put gives its holder the right, but not the obligation, to sell an underlying asset for the strike price K.

Your turn

For a put with $K = 100$, determine the payoff when

$$S_T = 70, \qquad S_T = 100, \qquad S_T = 130.$$

S_T	**Exercise?**	**Payoff**
£70	Yes	£30
£100	Indifferent	£0
£130	No	£0

$$P_T = \max(K - S_T, 0).$$

Payoff Is Not the Same as Profit

Long-call payoff

$$\text{Payoff}_T = \max(S_T - K, 0).$$

Long-call profit

Ignoring interest on the initial premium,

$$\text{Profit}_T = \max(S_T - K, 0) - c_0,$$

where c_0 is the option premium.

Break-even

For a call with $K = 100$ and premium $c_0 = 8$, at what underlying price does the buyer break even?

$$S_T = K + c_0 = 108.$$

When May the Option Be Exercised?

European option

May be exercised only on its expiration date.

American option

May be exercised at any time up to and including its expiration date.

Concept check

An American option is otherwise identical to a European option.
Should the American option ever be worth less?

No. It provides at least the same exercise opportunities and possibly more.